

SKSP: Selected Kernel Spatial Pattern Analysis in Euclidean and Hilbert Space for Decoding Motor Imagery

1st Zhang Zhibin
School of Computer Science and Engineering
Beihang University
Beijing, China
zhangzhibin@buaa.edu.cn

2nd Jianwei Niu
School of Computer Science and Engineering
Beihang University
Beijing, China
niu Jianwei@buaa.edu.cn

3th Lu Ren
School of Computer Science and Engineering
Beihang University
Beijing, China
by1706108@buaa.edu.cn

4th Zhenchao Ouyang
Zhongfa Aviation Institute
Beihang University
Hangzhou, China
ouyangkid@buaa.edu.cn

5th Shasha Mo*
School of Cyber Science and Technology
Beihang University
Beijing, China
moshasha@buaa.edu.cn

Abstract—Brain-computer Interface (BCI) has promising prospects as the most active research direction in the fields of brain science and artificial intelligence. The Common Spatial Pattern (CSP) algorithm has attracted extensive attention and has been demonstrated to be effective for Motor Imagery (MI) tasks. However, CSP is limited to capturing complex nonlinear characteristics under its linear assumption. Many variants of CSP cannot deal with multiple relationships simultaneously. This paper proposes a Selected Kernel Spatial Pattern (SKSP) algorithm for human behavior recognition. Electroencephalogram (EEG) signals are projected onto a high-dimensional Reproducing Kernel Hilbert Space (RKHS) by the selected kernel function, where the nonlinear relationship is transformed into a linear one. This approach exploits wavelet transform and CSP patterns to compute the energy spectrum and differential entropy of data in RKHS to represent EEG patterns of brain activities. In addition, a multi-view module was proposed to strengthen pattern representations, which employs mutual information maximization to select the optimal sliding window in both frequency and time domains. The experiments are conducted on the BCI Competition IV-2a dataset. The four classes' classification accuracy reaches 92% in subject seven. The proposed method outperforms the best state-of-the-art classification method by 4% in average subject accuracy.

Keywords—Kernel, Nonlinear, Common Spatial Pattern, Motor Imagery

I. INTRODUCTION

Electroencephalograph (EEG) is the most popular method to monitor brain electrical activities because of its non-invasive, low equipment cost, and high temporal resolution. As a non-stationary and low signal-to-noise ratio bioelectric signal, EEG is utilized in Brain-computer Interface (BCI) systems to restore

and enhance impaired hearing, vision, and motor abilities such as drones, wheelchairs, and prostheses. MI task activates neurons on the sensorimotor cortex and leads to an EEG pattern called Event-Related Desynchronization (ERD). The most widely used algorithm, Common Spatial Pattern (CSP), decomposes raw EEG signals into linear spatial patterns to maximize the variance difference across categories, which has been proven effective in capturing the ERD patterns [3]. Indeed, a significant number of CSP variants have been proposed and exhibited high performance [5]. Nonetheless, CSP only provides a partial and incomplete representation of EEG signals, given its limited field of view, strictly linear presumption, and loss of temporal information.

Due to the nonlinear composition of multiple brain source signals and noise sources, the EEG signal recorded on the scalp cannot be accurately analyzed without considering the inherent nonlinear relationship [4]. Along with deepening theoretical research, kernel methods have been introduced in CSP to capture the nonlinear relation of EEG signals. According to Cover's theorem [2], kernel methods embed the input data from the original space into a high-dimensional space via a nonlinear function in which the patterns can be considered linear relations. Some studies utilize a discriminative filter bank CSP combined with Kernel Principal Component Analysis (KPCA) and Structured Support Vector Machine (SSVM) to detect motor imagery movement patterns [14]. While KPCA and CSP share computational processes, resulting in partial overlap and significant computational costs. Despite the increasing computational complexity, KPCA performs worse on linear structured data than Principal Component Analysis (PCA). The linear and nonlinear relationships in EEG signals are equally important, and only considering one is inappropriate.

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Although the kernel CSP is a promising approach, it has been found to perform worse than the original CSP in some cases. This may be attributed to the fact that linear relationships are more important and discriminative. Additionally, EEG patterns differ among individuals, making a kernel function fitted for a specific subject unsuitable for others. This paper proposes a Selected Kernel Spatial Pattern (SKSP) that can fully utilize the multi-view information of EEG and recognize motor imagery patterns. The kernel functions are selected where both linear and nonlinear relationships are considered, and the EEG signals are projected onto high-dimensional Reproducing Kernel Hilbert Space (RKHS) by selected kernel functions. Consequently, nonlinear relationships are transformed into linear relationships. Then multiple sliding windows in the time domain and frequency domain are utilized to divide data into multiple segments. Mutual information serves as the criterion for selecting the optimal sliding window in both frequency and time domains, which can reduce parameters and provide guidance for understanding MI patterns. Furthermore, this approach combines and improves KPCA and CSP methods based on their partial similarities, resulting in a reduction in computational complexity. After window selection, spatial features can be extracted easily from the optimal segment of data in the feature space, and nonlinear relationships are well respected. On the BCIIV 2a dataset, the performance of SKSP is better than other kernel CSP methods.

The contributions of this paper can be summarized as follows:

- This paper proposes a Selected Kernel Spatial Pattern method that simultaneously captures nonlinear and linear relationships in the MI-BCI task. The selected kernel method converts the nonlinear relationships of the original space into linear relationships by projecting the data to a high-dimensional RKHS and then performing linear features. In this way, more information from different views of the data is obtained. To our knowledge, this is the first algorithm
- This paper augments the data set by designing sliding time windows and dividing multiple frequency bands. At the same time, we employ the correlation coefficient to select the optimal time window and frequency band to improve the classification accuracy.
- The experiments are conducted on the BCI Competition IV-2a dataset. The four classes' classification accuracy reaches 92% in subject seven. The proposed method outperforms the best state-of-the-art classification method by 4% in average subject accuracy.

The rest of this paper is organized as follows. Section II reviews the EEG signal processing methods and classification algorithms for motor imagery tasks. Section III presents the Selected Kernel Spatial Pattern method and the EEG-based convolutional classifier. In Section IV, this paper evaluates the proposed algorithm with the BCI competition IV-2a dataset. Finally, the advantages and disadvantages of the proposed algorithm are summarized in Section VI.

II. RELATED WORK

CSP is a spatial filtering feature extraction algorithm that extracts spatial distribution components from multi-channel BCI data for classification tasks. The basic principle of the CSP algorithm is to find optimal spatial filters for projection by diagonalizing a matrix to maximize the inter-class variance and minimize the intra-class variance. This approach leads to a higher eigenvector of discrimination. Multiple improvement methods have been designed such as Sliding Window CSP [20], Regularized CSP [6], and Probabilistic CSP [8].

Hekmatmanesh *et al.* [14] proposed a discriminative filter bank CSP method utilizing Kernel Linear Discriminant Analysis (KLDA), KPCA, and SSVM which performs well in some instances. Yimin Hou, *et al.* [15] proposed a novel scout-based Convolutional Neural Network (CNN) classifier for the classification of motor imagery tasks. In their system, the whole 64-channel EEG is divided into 10 scouts (half on the left and right). Then, features are extracted based on the morlet wavelets of the 10 scouts and are translated into a frequency domain for the later CNN.

Sakhavi modified the filter-bank common spatial patterns method to provide a new temporal representation and the CNN is designed and optimized called Channel-wise CNN (CW-CNN) and Channel-wise Convolution with Channel Mixing (C2CM) [10].

Inspired by Linear Discriminant Analysis (LDA) of DeepLDA [9], the DeepCSP [18] combines graph convolution neural network and common spatial patterns for EEG-based MI classification task. The CSP helps achieves accurate spatial feature extraction by maximizing the ratio of the between and within class scatter during CNN model training. However, the DeepCSP adopts 1D convolutional layers to directly deal with raw EEG and the lack of use of spatial information.

An RNN-based encoder-decoder framework [11] that is modified based on a modern machine-translation algorithm is proposed to transform the Electroencephalogram (EEG) of a person to corresponding spoken sentences. Although their study only explored restricted language with 30-50 unique sentences, their model is proved to have a high translation rate on a high- γ ECoG array with 120-250 electrodes.

A novel multi-channel Wiener filter (MWF)-based generic EEG artifact removal algorithm [17] is proposed. By using the signal correlation between EEG multi-channels, the filter can avoid being affected by different interference (i.e., patient, environment, or equipment) specific artifacts. The Signal-to-Noise Ratios (SNRs) are introduced to evaluate the performance of the filter based on the EEG signals before and after filtering.

CSP and deep learning methods have been combined to solve seizure prediction [12]. GCNs-Net [16] aims at precise decoding of brain activity measured by EEG. Considering there is a topological relationship between electrodes and signals, the graph convolutional neural networks (GCNs) is presented to decode the EEG signal by mapping them into a fixed Euclidean structure. They also analyzed the correlation

between EEG electrodes based on the PPC matrix and graph Laplacian.

Hamidreza *et al.* [13] fused the EEG signal and functional near-infrared spectroscopy (fNIRS) data and proposed a novel Recurrent-CNN model that incorporates the spatial relationship between the two data sources. The EEG electrodes are placed and divided based on a 10-5 system as four main parts in their EEG-fNIRS hybrid BCI system. The RCNN model combines normal CNN and long-short-term memory (LSTM) modules to learn spatial and temporal characteristics, respectively.

In [19], the authors present an energy-based EEG topographical representation method to generate high signal-to-noise ratio patterns. Three different energy mean constants and six different energy magnification algorithms are designed to achieve multi-object energy spectrum recognition. Based on this operation, the characteristics of the EEG signal are more prominent, and subsequent deep learning models can be easier to learn and converge. Rui Li *et al.* present a masked autoencoder based on a CNN-Transformer structure with self-supervised learning to deal with the abnormal circumstance of partially damaged signals [21]. And Xujin Li *et al.* proposed a Transformer framework with a temporal-spatial and frequency-spatial view [22].

It can be found that the early EEG analysis was mainly based on the time-domain characteristics of multi-channel linear data or directly used wave analysis for raw data. On the one hand, with the development of deep learning, models such as CNN and GCN have been introduced into EEG data, considering the linear relationship in the spatial domain. On the other hand, recent researches partially capture the nonlinear spatial relationship by KPCA. While both of them should be respected, there is no model that considers both linear and nonlinear relationships currently.

III. METHOD

This paper proposes a novel Selected Kernel Spatial Pattern algorithm with a correlation-based time window and frequency band selection module for human behavior recognition. The whole model can be divided into three stages: a generalized pre-processing stage, a selected kernel spatial pattern extraction stage, and a correlation-based sliding window selection stage. Firstly, EEG signals are preprocessed through several steps such as EOG removal, baseline correction, and bad epochs removal. Secondly, the proposed kernel functions project the preprocessed EEG data into high-dimensional RKHS where nonlinearity is transformed into linearity. Then a sliding window module is proposed that EEG data is divided into multiple pieces in both the time domain and frequency domain (Wavelet Transform is used). Specifically, optimized EEG signals in the most appropriate time window and frequency band are selected based on correlation analysis and performance evaluation. Finally, Common Spatial Pattern is used as the feature extraction algorithm for human behavior recognition, and the whole workflow is presented in Fig. 1.

A. Pre-processing

Considering the EEG collector used in this study contains multiple channels, and the sampling results often contain a lot of noise due to differences in environment and subjects, dealing with noisy and dirty EEG signals is a challenge. The first step of dealing with the signals is to generate stable and clear data from the raw labeled EEG data.

The frequency band selection uses a medium band pass filter to get signals within 8-30 Hz for the MI task, where μ -band is 8-13 Hz and β -band is 13-30 Hz. According to [7], the EEG signals generated during MI tasks, the energy of μ -band (8-13 Hz) in the motor cortex of the brain was observed to decrease, and the energy of β -band (13-30 Hz) was observed to increase, respectively. The former phenomenon is called ERD, and the latter is called event-related synchronization (ERS). After noise reduction, the remained ERD and ERS attributes can help improve the accuracy of MI task classification tasks.

1) *EOG removal.*: The ElectrooculoGraphy (EOG) from the movement of the eye muscles also has an impact on the EEG, so some data sets collect additional EOG signals to assist in noise analysis. When dealing with EEG with EOG, the Independent Component Analysis (ICA) is considered a blind source separation algorithm to reduce EOG components in EEG.

$$Epoch(i) = data_{t,n}, event_i \quad (1)$$

2) *Baseline correction.*: To discriminate the EEG signal corresponding to the motor imagery from all the data acquired, the processed data is divided into several segments called Epochs, as Eq. 1. Where $data_{t,n}$ donates the raw data, t is the period of $[i - 0.2s, i + 1.5s]$ at the i -th event, n is the number of the channel, and $event_i$ is the i -th labeled event. Baseline correction is used to further eliminate the interference of spontaneous EEG signals (not related to MI tasks). For every single channel, the mean value of the signal 200 ms before the event is subtracted from the signal value in the epoch.

3) *Bad epochs removal.*: The amplitude of EEG is usually in the range of a few micro-volts to tens of micro-volts. When the amplitude of a signal in an epoch is greater than 100, other stronger signals cover the corresponding EEG signal we need. Therefore, the related bad epochs need to be eliminated.

B. Time window and frequency band selection

This paper proposes a sliding window module to select the optimal frequency bands and time windows thus the whole algorithm can achieve the best performance. Correlation facilitates the detection of active cortical areas associated with MI tasks. Studies have shown that when subjects perform MI tasks, those channels related to MI tasks should contain common information, and there is a correlation between such channels across multiple MI tasks.

This paper evaluates the influence of time windows and frequency bands on classification accuracy for CSP. Some studies focused on the influence of frequency bands and

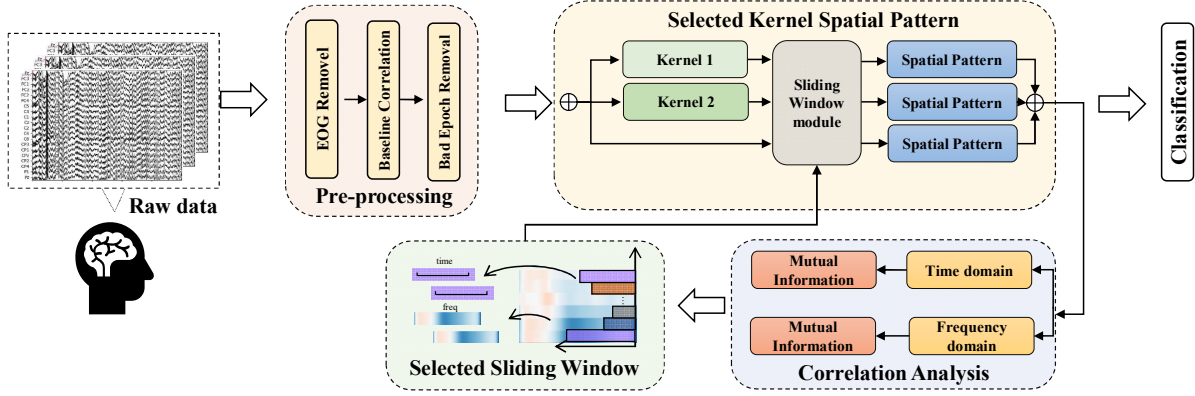


Fig. 1: The overall process of our proposed Selected Kernel Spatial Pattern for human behavior recognition.

divided them into several unduplicated sub-bands (delta:0-4Hz, theta:4-8Hz, alpha:8-13Hz, beta:13-30Hz).

But they ignored potential connections between different subbands. Therefore, this paper designs 43 unique sub-bands with different widths and starting points. This idea is applied to time windows and 11 time windows with different lengths are designed. EEG data is divided into multiple epochs with specific frequency bands and time periods, and SKSP is utilized to extract the features. Then this paper calculates the mutual information and classification accuracy for different frequency bands and time windows respectively. According to the result, we select the certain frequency band and time window with the best performance as guidance. Finally, the data within the specified optimal time window and frequency band are used for subsequent processing.

C. SKSP

The SKSP module consists of two parallel parts, one is the linear projection part and the other is the nonlinear projection part. The linear projection part is the same as CSP which diagonalizes the matrix to maximize the inter-class variance and minimize the intra-class variance to obtain discriminative patterns. Referring to KPCA's approach, the nonlinear projection part utilizes kernel functions to project data into high-dimensional Reproducing Kernel Hilbert Space where nonlinear relationships are respected. The EEG dataset is given as $\mathbf{X} \in \mathcal{R}^n$. For the selected kernel function Φ , the metadata space is projected into the high-dimensional space $\mathcal{H}^s$ where $s > n$.

$$\Phi: \mathcal{R} \rightarrow \mathcal{H} \quad (2)$$

In this study, the nonlinear map is the radial basis function kernel. The average mixed spatial covariance matrix is computed:

$$\mathbf{C}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \mathbf{C}_i^n \quad (3)$$

$$\mathbf{C}_i^n = \frac{\mathbf{X}_i \mathbf{X}_i^T}{\text{trace}(\mathbf{X}_i \mathbf{X}_i^T)} \quad (4)$$

where $\mathbf{C}_i$ is the covariance matrix of each class of EEG data and i represents the index of the MI categories. Trace $\mathbf{X}$ is the sum of the diagonal elements of $\mathbf{X}$ and $\mathbf{X}^T$ represents the transpose matrix of $\mathbf{X}$. The covariance matrix $\mathbf{C}_i$ is normalized and averaged in class i , where N_i is the number of trials of the i -th type of motor imagery movement.

Then, the eigenvalue decomposition is performed on the mixed spatial covariance matrix $\mathbf{C}$.

$$\mathbf{C} = \mathbf{C}_1 + \mathbf{C}_2 = \mathbf{U}_c \lambda_c \mathbf{U}_c^T \quad (5)$$

where $\mathbf{U}_c$ is the eigenvector matrix of the mixed space covariance matrix $\mathbf{C}$, and the λ_c is a diagonal matrix formed by corresponding eigenvalues arranged in descending order. The whitening matrix $\mathbf{P}$ is:

$$\mathbf{P} = \sqrt{\lambda_c^{-1}} \mathbf{U}_c^T \quad (6)$$

The average space covariance matrix of each class is transformed by whitening matrix as follows:

$$\begin{aligned} \mathbf{S}_1 &= \mathbf{P} \mathbf{C}_1 \mathbf{P}^T = \mathbf{B} \mathbf{A} \mathbf{B}^T \\ \mathbf{S}_2 &= \mathbf{P} \mathbf{C}_2 \mathbf{P}^T = \mathbf{B} (\mathbf{E} - \mathbf{A}) \mathbf{B}^T \end{aligned} \quad (7)$$

where $\mathbf{E}$ is the unit matrix, and the eigenvector corresponding to the maximum eigenvalue of $\mathbf{S}_1$ is the eigenvector corresponding to the minimum eigenvalue of $\mathbf{S}_2$. And the eigenvector corresponding to the minimum eigenvalue of $\mathbf{S}_1$ is the eigenvector corresponding to the largest eigenvalue of $\mathbf{S}_2$. The projection matrix $\mathbf{W}$ is reconstructed:

$$\mathbf{W} = \mathbf{B}^T \mathbf{P} \quad (8)$$

where matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$. Then select the first and last M columns from the projection matrix $\mathbf{W}$ to construct the optimal spatial filter $\mathbf{W}^* = [W_1, \dots, W_M, W_{N-M}, \dots, W_N]$. The

four classes of EEG signals are filtered by the optimal spatial filter $\mathbf{W}^*$, and the feature $\mathbf{f}_m$ is extracted from the filtered EEG signal:

$$\mathbf{f}_m = \log(\text{var}(\mathbf{W}_m^{*T} X / \sum_{m=1}^{2M} \text{var}(\mathbf{W}_m^{*T} X)) \quad (9)$$

According to Mercer's theorem [1], any function can serve as a kernel function if it satisfies positivity and symmetry:

$$\int \int f(x) K(x, y) f(y) dx dy \geq 0 \quad (10)$$

$$K(x, y) = K(y, x)$$

While a binary function $K(x, y)$ can be regarded as an infinite-dimensional matrix, similar to matrix eigenvalues and eigenvectors, there exist eigenvalues λ and eigenfunctions $\psi(x)$ such that

$$\int K(x, y) \psi(x) dx = \lambda \psi(y) \quad (11)$$

One kernel function corresponds to infinite eigenvalues and infinite orthogonal characteristic equations. And eigenvalue decomposition can also be performed on the kernel method:

$$K(x, y) = \sum_{i=1}^{\infty} \lambda_i \psi_i(x) \psi_i(y) \quad (12)$$

where $\langle \psi_i, \psi_j \rangle = 0, i \neq j$. $\{\psi\}_{i=1}^{\infty}$ constitutes a set of orthogonal bases in the function space.

We use $\{\sqrt{\lambda_i} \psi\}_{i=1}^{\infty}$ as a set of orthogonal bases to construct a Hilbert space $\mathcal{H}$ where any vector can be represented as a linear combination of the set of bases. So function f can be expressed as an infinite-dimensional vector in $\mathcal{H}$: $f = [f_1, f_2, \dots]^T$. For a given kernel function ϕ , the input data space $\mathcal{R}^n$ is projected into the high-dimensional feature space $\mathcal{R}^m$, ($m > n$):

The average mixed spatial covariance matrix in Hilbert space $\mathcal{H}$ is constructed by

$$\mathbf{C}_t^{\Phi} = \frac{1}{N} \sum_{j=1}^N (\Phi(x_j) (\Phi(x_j))^T) \quad (13)$$

When directly performing the inner product calculations in high-dimensional space, there is a really high computational complexity. In this case, the algorithm can be effectively implemented in the input space through kernel tricks to avoid this difficulty.

$$\Phi(\mathbf{x}) = K(x, \cdot) = (\sqrt{\lambda_1} \psi_1(x), \sqrt{\lambda_2} \psi_2(x), \dots)^T \quad (14)$$

where ϕ represents a vector in feature space $\mathcal{H}$. Therefore, we do not need to know what the mapping function is and where the feature space is. As long as it is a symmetric positive definite function, there must be a mapping and feature so that

$$\langle \Phi(\mathbf{x}), \Phi(\mathbf{y}) \rangle_{\mathcal{H}} = \langle K(\mathbf{x}, \cdot), K(\mathbf{y}, \cdot) \rangle_{\mathcal{H}} = K(\mathbf{x}, \mathbf{y}) \quad (15)$$

We denote $\phi = [\Phi(x_1), \dots, \Phi(x_N)]$ and eigenvalue decomposition is performed $\Phi = \mathbf{U}^T \mathbf{U}$. The orthonormal eigenvectors γ

Algorithm 1: The process of the Selected Kernel Spatial Pattern Algorithm.

Input: The EEG source data $\mathbf{X}$

```

1 Initialize kernel function  $\Phi$ ,  $\mathbf{t}_m$ ,  $\text{freq}_n$ ;
2 while  $\mathbf{t}_m \in \text{time}$  do
3   separate data  $\mathbf{X} \rightarrow \mathbf{X}_t$ ;
4   while  $\text{freq}_n \in \text{frequency}$  do
5     separate data  $\mathbf{X} \rightarrow \mathbf{X}_f$ ;
6     if  $\Phi$  is linear then
7       Calculate the average mixed spatial covariance
8        $\mathbf{C}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \frac{X_i X_i^T}{\text{trace}(X_i X_i^T)}$ ;
9       eigenvalue decomposition  $\mathbf{C} = \mathbf{U}_c \lambda_c \mathbf{U}_c^T$ ;
10      whitening matrix  $\mathbf{P} = \sqrt{\lambda_c^{-1}} \mathbf{U}_c^T$ ;
11      Transform  $\mathbf{C}_i$ :  $\mathbf{S}_i = \mathbf{P} \mathbf{C}_i \mathbf{P}^T$ ;
12       $\{\mathbf{S}_i\} \leftarrow \max(\mathbf{B} \mathbf{A} \mathbf{B}^T)$ ;
13      reconstruct projection matrix  $\mathbf{W} = \mathbf{B}^T \mathbf{P}$ ;
14      feature  $\mathbf{f}_m = \mathbf{W}^T \mathbf{X}$ ;
15     else if  $\Phi$  is nonlinear then
16       Project input data to RKHS  $\Phi: \mathcal{R} \rightarrow \mathcal{H}$ ;
17       Calculate the average mixed spatial covariance
18        $\mathbf{C}_t^{\Phi} = \frac{1}{N} \sum_{j=1}^N (\Phi(x_j) (\Phi(x_j))^T)$ ;
19       eigenvalue decomposition  $\mathbf{C}_t^{\Phi} = \mathbf{U} \lambda_j \mathbf{U}^T$ ;
20       whitening matrix  $\mathbf{Y} = \sqrt{\lambda_j^{-1}} \mathbf{U}^T$ ;
21       Transform  $\mathbf{C}_t^{\Phi}$ :  $\mathbf{S}_j = \mathbf{Y} \mathbf{C}_t^{\Phi} \mathbf{Y}^T$ ;
22        $\{\mathbf{S}_j\} \leftarrow \max(\mathbf{D} \mathbf{A} \mathbf{D}^T)$ ;
23       reconstruct projection matrix  $\mathbf{W}' = \mathbf{D}^T \mathbf{Y}$ ;
24       feature  $\mathbf{f}_n = \sqrt{\lambda_j^{-1}} \gamma^T \mathbf{U} \Phi(x)$ 

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Output: $\mathbf{f}_m, \mathbf{f}_n$;

of $\mathbf{U}$ correspond to the m largest eigenvalues λ_j . The whitening matrix $\mathbf{Y}$ is similar to the equation 6. After whitening and the other same process, we can obtain the feature transformed by SKSP:

$$\mathbf{f}_n = \sqrt{\lambda_j^{-1}} \gamma^T \mathbf{U} \Phi(x) \quad (16)$$

$$= \sqrt{\lambda_j^{-1}} \gamma^T [k(x_1, x), k(x_2, x), \dots, k(x_m, x)]$$

IV. EXPERIMENTS

A. Dataset

Our proposed method and the compared methods are validated on BCI Competition IV Dataset 2a. This dataset records the EEG data of 4 types of motor imagery tasks of nine healthy subjects. Each subject is required to complete 288 experiments in the four types of motor imagery tasks on the left hand, right hand, feet, and tongue. 22 channels were recorded with a sample rate of 100 Hz. Each experiment process: First, a brief prompt beep appears ($t=0$), and a fixed cross is displayed in the center of the computer screen. After two seconds ($t=2s$), a left, right, up, or down arrow appears on the computer screen, indicating the subject to perform four categories of left hand, right hand, feet, and tongue movements. A black screen appears within 6-8 s, and the time axis of a single experiment is shown in Figure 5.

There is an apparent ERD phenomenon in the frequency band (13-30 Hz), so this paper intercepts the EEG data 0.5-4 s after the visual cue in each experiment and uses a finite

unit impulse response filter (Finite Impulse Response, FIR) to analyze the EEG signal. A band-pass filter of 4 to 40 Hz is performed.

B. Classification performance

TABLE I: Classification accuracies of proposed model and baselines in BCIIV 2a dataset.

	FBCSP	CW-CNN	C2CM	Proposed
subject1	76.00	86.11	87.50	90.97
subject2	56.5	60.76	65.28	66.57
subject3	81.25	86.81	90.28	88.74
subject4	61	67.36	66.67	73.05
subject5	55	62.50	62.50	64.72
subject6	45.25	45.14	45.49	60.14
subject7	82.75	90.62	89.58	92.10
subject8	81.25	81.25	83.33	84.20
subject9	70.75	77.08	79.51	84.23
Average	67.75	73.07	74.46	78.30

We utilize the SKSP algorithm in both 43 frequency bands and 11 time windows of EEG signals to find the optimal frequency-temporal region. To maximize the utilization of spatial information inherent in the data, SKSP captures nonlinear features in high-dimensional space and linear features in low-dimensional space simultaneously. After pre-processing, metadata space is projected into RKHS by radial basis function and the Common Spatial Pattern is used for feature extraction. To prevent matrix irreversibility during the diagonalization process in CSP, we add 0.0001 to each element on the matrix's main diagonal. The number of projection filters is 24 as a hyperparameter in our model.

Considering the individual variability of EEG signals, we train the model separately for each subject. Table I shows the four classification accuracies of our method achieve state-of-the-art results in comparison with other baseline methods on BCIIV 2a datasets, which indicates that the nonlinear relationship can improve the efficiency of the model, especially on the EEG signals with complex patterns.

Specifically, the classification accuracy of our model reaches 92% and the average performance reaches 78.3% for the best-performing subject. The performance of SKSP is significantly higher than those compared methods.

The differences in classification accuracy among different subjects may be attributed to diversity in brain activity patterns. Different mental states and experimental environments during data acquisition could also contribute to these.

C. Ablation Study

Our SKSP method performs the parallel computation of different features in both the original and high-dimensional space and combines them. To demonstrate the effect of parallel computation in both the original and high-dimensional space, we implement the ablation study by only extracting linear

or nonlinear relationships. The first four columns of Table II show the classification accuracy by only computing the nonlinear features in the high-dimensional space. Classification accuracy by only computing the linear features in the source space is shown in the three columns in the middle of Table II. Therefore we can evaluate whether projecting to RKHS by kernel functions is better for the CSP algorithm to capture discriminative patterns from MI tasks. Obviously, the classification accuracy reduces after adding the kernel method. The possible reason could be that the distribution of the EEG data is mostly linear, with only a small portion being nonlinear. In this case, extracting only nonlinear features resulted in a decrease in classification accuracy. In addition, the performance of the kernel method is usually affected by the choice of parameters, and inappropriate parameter selection may also lead to a decrease in accuracy.

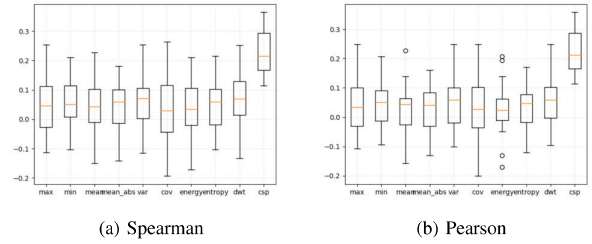


Fig. 2: Boxplots of coefficient comparisons of six features

The Pearson correlation coefficient is applicable to continuous data with a linear relationship between two variables. As long as the observed values of two variables are paired or derived from continuous variable observations, the Spearman correlation coefficient can be used for analysis regardless of the overall distribution form of the two variables and the sample size. Correlation coefficients are utilized to measure the effectiveness of several feature extraction methods on EEG signals in Fig. 2. CSP performs the best compared to other statistical characteristic.

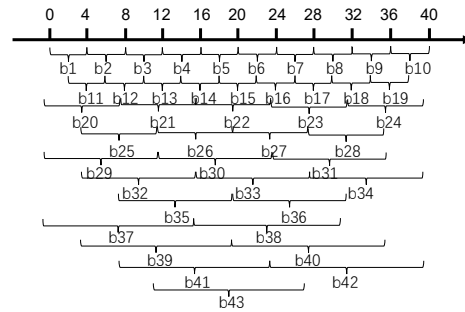


Fig. 3: Schematic diagram of frequency band division.

TABLE II: Classification performance of the proposed method with different frequency bands and different components on BCIV 2a dataset.

	KPCA				PCA			KPCA+PCA				
	8-14	14-30	8-30	4-40	8-14	14-30	8-30	8-14	14-30	8-30	8-30(h)	4-40
subject1	49.47	32.38	46.98	49.82	67.62	81.49	86.12	82.47	60.87	80.90	90.97	84.69
subject2	25.09	27.56	28.62	27.21	29.68	57.60	58.66	62.53	64.74	66.32	66.57	65.94
subject3	51.28	36.63	49.08	44.69	77.29	73.63	84.62	95.50	74.29	91.30	88.74	85.22
subject4	41.67	37.38	40.35	39.47	40.79	68.86	64.91	67.87	62.53	66.03	73.05	64.57
subject5	25.72	27.54	27.54	28.62	35.51	51.81	49.28	67.93	69.52	68.66	64.72	70.61
subject6	33.02	29.77	38.60	35.35	43.72	42.33	52.56	53.08	47.35	60.12	60.14	55.73
subject7	33.57	34.66	37.91	41.88	48.38	83.03	79.42	82.84	84.03	88.38	92.10	94.21
subject8	42.07	34.69	44.28	44.28	85.61	74.54	89.67	74.00	63.43	78.61	84.20	77.56
subject9	49.62	54.55	62.12	58.33	84.09	77.27	82.20	63.63	73.39	85.74	84.23	78.14
average	39.06	35.01	41.72	41.07	56.96	67.84	71.94	72.21	66.68	76.23	78.30	75.19

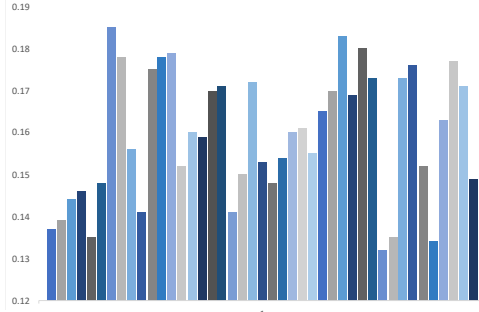


Fig. 4: Mutual Information values of features extracted in 43 frequency bands.

D. Selections of frequency bands for CSP

The individual classification accuracies for all subjects with different frequency bands were listed in Fig. 4.

We use sliding windows of 5 different scales to divide the frequency of the EEG signal. The sliding step size is 2Hz for windows with a scale of 4Hz, and 4Hz for windows with a scale of 8Hz, 12Hz, and 16Hz. We designed a total of 43 sliding frequency band windows.

We investigated the mutual information value for each frequency band of features computed by CSP in different frequency bands. We could find that CSP obtains the highest mutual information value of 0.19 at the 7th band. The mutual information values of the five three frequency bands are less than 0.15. It reveals that the δ band and θ band could provide little contribution to the accuracy.

E. Selections of time windows for CSP

The specific process of experimental paradigm Setting is shown as a timeline in Fig. 5. The cue occurs in the second to third seconds ($t=2s-3s$), and motor imagery is considered to occur in the third second ($t=3s$). We assume that the signal within 0.5 seconds prior to MI occurrence contains useful information ($t=2.5s-3s$). Based on the assumption, 11 time windows are designed from $t=2.5s$ to $t=6s$ within different lengths.

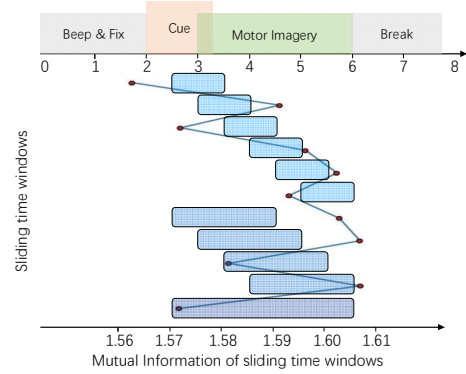


Fig. 5: Mutual Information values of feature extracted in 11 selected time windows.

We investigate the mutual information value for each time window of features computed by CSP. We could find that CSP obtains the highest mutual information value of 1.61 at the 10th time window. Therefore the EEG signals within one second after stimulation provided the greatest contribution. This indicates that reasonable time window selection helps to extract discriminative features, and variations within a limited time range have a negligible effect on the results.

V. CONCLUSION

This study proposes a Selected Kernel Spatial Pattern (SKSP) method with a sliding window module to capture discriminative patterns in the MI-BCI task. Our method takes full advantage of EEG signals by exploring nonlinear and linear relationships simultaneously through the multiview RKHS and normal space structure. SKSP is validated on the BCIV-2a dataset including nine subjects and 2592 trials. The experimental results show that our method achieves significantly higher performance than various advanced baseline methods. The result demonstrates that the proposed SKSP method can learn multi-view EEG representations from the frequency and time domains.

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